

Practice Quiz: Communication — Signals Through the Mesh

Part A: Multiple Choice

1. In a filet crochet mesh, solid blocks of double crochets transmit force differently than chain spaces. In neural network terms, this difference is best described as: A) Different numbers of layers in the network B) Different connection weights — solid blocks have high weights (strong signal transmission) and chain spaces have low weights (attenuated signal) C) Different input values entering the network D) Different output labels produced by the network
2. When a structural error in Row 3 of a mesh pattern causes misalignment that propagates into Rows 4, 5, and 6, this is most analogous to: A) Forward propagation of input through network layers B) Error propagation through a network, where an upstream mistake distorts downstream outputs C) Dropout regularization in a neural network D) The vanishing gradient problem
3. A filet crochet chart that shows solid and open blocks in a grid is best described as: A) A purely decorative illustration with no formal structure B) A graph representation of the fabric network — literally a network diagram drawn on paper C) An audio encoding of the pattern D) A random arrangement of symbols
4. Color pooling in variegated yarn occurs because: A) The yarn manufacturer designs the argyle pattern into the yarn B) The stitch pattern alone creates the color effect regardless of yarn choice C) The visual pattern emerges from the interaction of the yarn's color sequence and the stitch placement — an emergent phenomenon of signal interacting with network structure D) The crocheter manually places each color at a specific location
5. When a crocheter pulls the working yarn and feels the tension propagate backward through the fabric, this is an example of: A) A signal flowing backward through the network, analogous to feedback in recurrent systems B) Forward propagation only C)

An event that has no parallel in network science D) Dropout, where connections are randomly removed

6. Crochet pattern notation (abbreviations, charts, stitch counts) functions as a communication protocol because: A) It is written on paper B) It includes standardized syntax (abbreviations, formatting), shared semantics (each symbol maps to a specific action), and is transmitted through a channel (book, PDF, screen) C) It is used only by professional designers D) It cannot be misunderstood
7. A crocheter who detects a stitch count error and frogs back several rows to fix it is performing an operation most similar to: A) Forward propagation in a feedforward network B) Backpropagation — tracing the error signal backward through the layers to its source and adjusting the "weights" (stitches) to correct it C) Adding a new layer to the network D) Changing the activation function

Part B: Short Answer

1. A filet crochet piece is 30 blocks wide and 45 rows tall. Each block is either solid or open. Describe this piece as a communication system: what is the "message," what is the "medium," and how much visual information (in binary pixels) does it encode? How does the gauge (stitches per inch) affect the "resolution" of this display?
2. Compare how a written crochet pattern and a stitch chart each represent the same fabric. Explain how the written pattern mirrors the sequential process of crocheting (one stitch after another) while the chart mirrors the spatial topology of the finished fabric (nodes in a grid). Which format makes it easier to trace signal paths through the mesh, and why?
3. You are crocheting a lace mesh shawl and gently tug on the bottom point. Describe what happens physically as the force propagates through the network of chain spaces and stitch posts. Where does the signal travel fastest? Where is it attenuated? How does this physical signal propagation illustrate the concept of connection weights in a neural network?